

Unit 2

6 warm-ups · 24 vocabulary terms

By the end of this unit, I can...

- ☐ I can build a color with `rgb(r, g, b)`.
- ☐ I can use opacity as a number from 0 to 100.
- ☐ I can create a variable and give it a value.
- ☐ I can explain what the `=` sign does.
- ☐ I can tell an integer from a float from a string.
- ☐ I can recognize a Boolean value.
- ☐ I can evaluate an arithmetic expression.
- ☐ I can use the operators `+` `-` `*` `/` `//` `%`.
- ☐ I can use a variable to position a shape.
- ☐ I can change one variable to move several shapes.
- ☐ I can choose clear variable names.
- ☐ I can find a simple bug in an expression.

How to use these pages

Each warm-up is two pages. Page 1 is the header, the START HERE strip (A–E), and Vocabulary. Page 2 is Part 1 (everyone) and Part 2 (keep going if you finish early). Mark your own or a partner's work — it does not count against you.

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 11 — Color by number

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can build a color with <code>rgb(r, g, b)</code> .
C	☐ got it ☐ shaky ☐ not yet I can use opacity as a number from 0 to 100.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	A goal — small is fine — for this class period:

Vocabulary — write each in your own words

rgb

gradient

opacity (0–100)

channel (r, g, b)

Part 1 — core work

```
fill = rgb(255, 140, 0)
```

Each rgb channel goes 0–255. What color is `rgb(0, 0, 0)`? What about `rgb(255, 255, 255)`?

A gradient blends two colors across a shape. Name two colors you would blend for a sunset.

PART 2 — keep going if you finish early.

Write an `rgb()` color for your favorite color, then explain what your three numbers do.

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 12 — Variables

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can create a variable and give it a value.
C	☐ got it ☐ shaky ☐ not yet I can explain what the = sign does.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	One word for how today is going so far:

Vocabulary — write each in your own words

variable

assignment (=)

value

name (identifier)

Part 1 — core work

```
score = 0  
name = 'Robo'
```

In `score = 0`, which side is the NAME and which side is the VALUE? What does the `=` actually do?

Create a variable called `lives` and set it to 3.

PART 2 — keep going if you finish early.

After `lives = 3`, you run `lives = lives - 1`. What is `lives` now, and why?

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 13 — Data types

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can tell an integer from a float from a string.
C	☐ got it ☐ shaky ☐ not yet I can recognize a Boolean value.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	Something you are looking forward to this week:

Vocabulary — write each in your own words

integer (int)

float

string

Boolean

Part 1 — core work

Label each value with its type — int, float, string, or Boolean: 7 · 3.5 · 'hello' · True

Why do we put quotes around a string but not around a number?

PART 2 — keep going if you finish early.

The value '7' and the value 7 look alike. How are they different to the computer?

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 14 — Expressions and operators

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can evaluate an arithmetic expression.
C	☐ got it ☐ shaky ☐ not yet I can use the operators + - * / // %.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	A person you could ask for help if you get stuck:

Vocabulary — write each in your own words

expression

operator

// (integer divide)

% (remainder)

Part 1 — core work

total = 3 * 4 + 2

Evaluate each expression: $3 * 4 + 2$ · $10 // 3$ · $10 \% 3$

Which happens first in $3 * 4 + 2$ — the * or the + ? Why?

PART 2 — keep going if you finish early.

Use % (remainder) to decide whether 17 is even or odd. What expression tells you, and what does it give?

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 15 — Variables that draw

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can use a variable to position a shape.
C	☐ got it ☐ shaky ☐ not yet I can change one variable to move several shapes.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	One thing that is going well outside of this class:

Vocabulary — write each in your own words

parameter

reuse

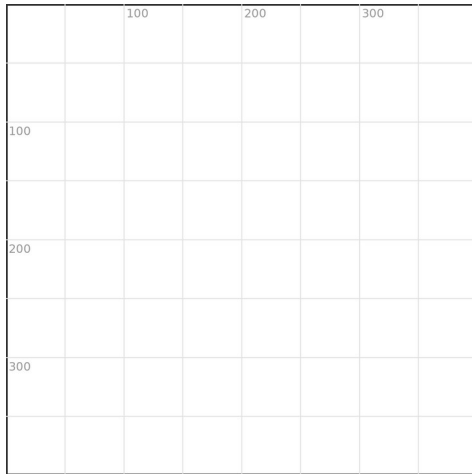
update

expression in a call

Part 1 — core work

```
x = 100  
Circle(x, 200, 30)  
Circle(x + 120, 200, 30)
```

Both circles use x . On the grid, plot them for $x = 100$. Then plot where they move if x becomes 180.



Why is using a variable for x better than typing the number twice?

PART 2 — keep going if you finish early.

Add a third circle that is always 120 to the right of the second one. Write its center x using the variable x .

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 16 — Naming and debugging

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can choose clear variable names.
C	☐ got it ☐ shaky ☐ not yet I can find a simple bug in an expression.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	If you finish early today, what will you try next?

Vocabulary — write each in your own words

identifier

readability

debugging

comment (#)

Part 1 — core work

Which name is clearer for a player's score — `s` or `score`? Why does it matter once the program gets long?

```
lives = 3
lives = lives - 1
```

This code has a typo bug. Find it and write the corrected two lines.

PART 2 — keep going if you finish early.

Add a comment (`#`) that explains what `lives = lives - 1` does in a game.
